

**SCSE** 

## **Stablecoin Clearing & Settlement Engine**

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*Product Owner | Fintech & Innovation  
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**Seeking Opportunities in Singapore**

I am looking for roles in **Product Management, Fintech, Payments, RegTech,** and **Digital Assets.**

"I am not just a Product person. **I build.**"

I have worked across product delivery, user research, and cross-agency collaboration. I enjoy solving complex problems and bringing structure to early ideas.

**I care deeply about building products that create real impact.**

# The Problem: Digital Money is Siloed

In my work analyzing **Cross-Border Payments & CBDCs**, I've seen how liquidity fragmentation kills efficiency.

1. ● **Trapped Liquidity:**

Stablecoins sit idle in different chains and wallets, requiring massive pre-funding.

2. ● **Slow Finality:**

Traditional blockchains (15s - 10 min) are too slow for high-frequency point-of-sale payments.

3. ● **High Gas Costs:**

Settling *every* coffee purchase on-chain is inefficient.

“**Goal:** Build a **Hybrid Infrastructure** that combines database speed with blockchain finality.

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# The Solution: SCSE

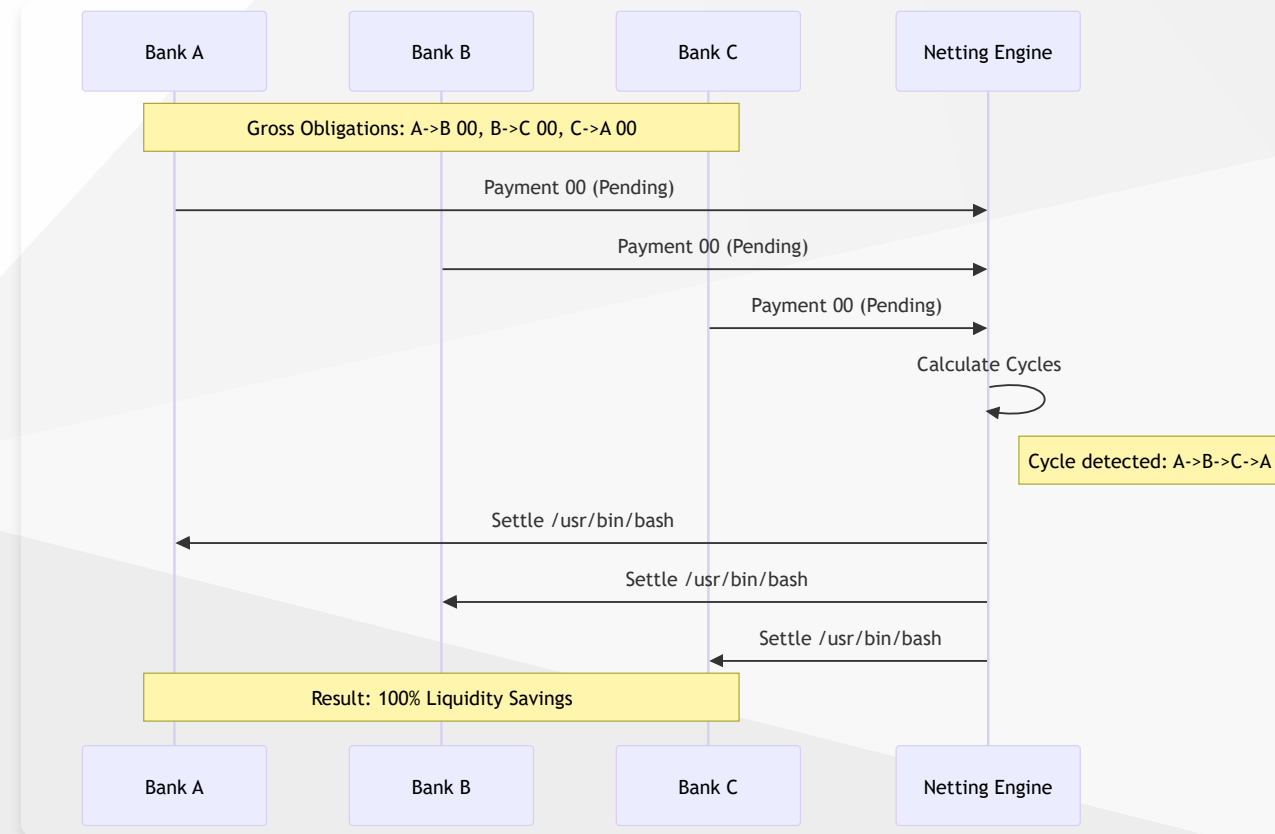
A portfolio-grade financial engine that implements **Real-Time Gross Settlement (RTGS)** and **Multilateral Netting** for Stablecoins.

## Hybrid Architecture

- ⚡ **Layer 1 (Off-Chain):**  
10k+ TPS. Immediate clearing. Privacy-preserving compliance.
- 💰 **Layer 2 (On-Chain):**  
Sepolia Testnet. Cryptographic proof of settlement.

## 100% Liquidity Savings

Algorithms identify and resolve circular debts (A->B->C->A).



# Technical Architecture

I built a full-stack simulation to demonstrate end-to-end payment lifecycles.

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- **Frontend:** React, Mantine, Zustand.
- **Web3:** Wagmi, Viem, Solidity (Hardhat).
- **Logic:** Python (FastAPI/SQLAlchemy) adapted to client-side logic for the demo.

# Engineering Philosophy & Impact

This project reflects my ability to bridge **Policy** and **Code**:

1. **Compliance-First:**

Built with hooks for **Travel Rule (IVMS101)** compliance, ensuring regulatory readiness.

2. **Standards-Based:**

Uses **ISO 20022** terminology (DebtorAgent, CreditorAgent) for interoperability.

3. **Real-World Reliability:**

Implements proper **Double-Entry Accounting** logic, not just simple token transfers.

# About the Builder

**Sivasubramanian Ramanathan**

*Product Owner | Fintech, RegTech & Digital Innovation*  
*PMP | PSM II | PSPO II*

I specialize in taking messy, real-world complexity and structuring it into reliable products.

**Open for roles that sit between policy, technology, and stakeholder engagement.**

## Lets Connect

I am ready to bring this level of engineering rigor and product thinking to your team.

-  **Portfolio:** [sivasub.com](https://sivasub.com)
-  **LinkedIn:** [linkedin.com/in/sivasub987](https://linkedin.com/in/sivasub987)
-  **Code:** [github.com/siva-sub/SCSE](https://github.com/siva-sub/SCSE)

**Live Demo:**

[siva-sub.github.io/Stablecoin-Clearing-and-Settlement-Engine](https://siva-sub.github.io/Stablecoin-Clearing-and-Settlement-Engine)